

Virtual Agent–Driven SLA Breach Awareness & Justification System

Team Members:

- ◆ Vaase Ammulu
- ◆ Akshitha pallem
- ◆ Kare sirisha
- ◆ Naidu Divya Bhavani
- ◆ Mothe Naga Sri Hema

2. Project Overview

Purpose

- Automate SLA breach awareness.
- Notify incident assignees before SLA breach.
- Collect SLA breach reasons.
- Store justification for audit.
- Improve SLA governance.

Key Features

- Incident Management
 - SLA Monitoring
 - Flow Designer Automation
 - Notifications
 - Virtual Agent Conversation
 - Dashboard & Reports
 - Audit Trail
-

3. Architecture

Instead of MERN architecture, use ServiceNow architecture:

```
User
|
Incident Creation
|
Incident Table
|
Task SLA
|
Flow Designer
|
Notification
|
Virtual Agent
|
Collect Justification
|
Incident Updated
|
Reports & Dashboard
```

4. Setup Instructions

Prerequisites

- ServiceNow Personal Developer Instance (PDI)
- System Administrator Role
- Flow Designer Plugin
- Notifications
- Reports & Dashboard
- Internet Browser

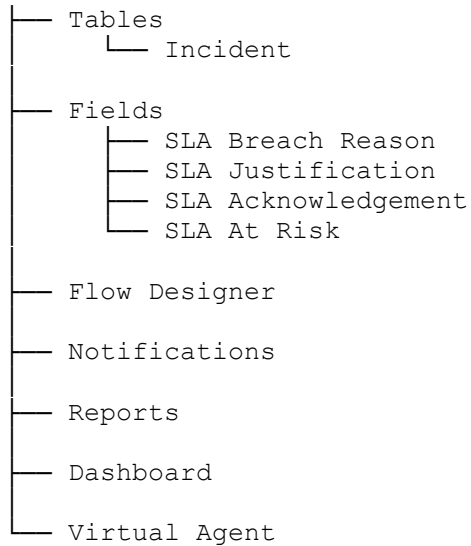
Installation

1. Log in to ServiceNow.
2. Create Application.
3. Create Incident Fields.
4. Configure SLA.
5. Create Flow Designer Flow.
6. Configure Notifications.
7. Create Reports.
8. Create Dashboard.

5. Folder Structure

Since ServiceNow does not use a client/server folder structure like MERN, replace it with application modules.

Application



6. Running the Application

1. Login to ServiceNow.
 2. Create an Incident.
 3. SLA starts automatically.
 4. At 80% SLA completion:
 - o Notification is sent.
 5. Virtual Agent requests justification (where supported).
 6. User submits reason.
 7. Dashboard updates automatically.
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7. API Documentation

Since ServiceNow provides REST APIs, include:

API	Method	Purpose
/api/now/table/incident	GET	Retrieve incidents
/api/now/table/incident	POST	Create incident

API	Method	Purpose
/api/now/table/task_sla	GET	View SLA records
/api/now/table/incident/{sys_id}	PATCH	Update justification

8. Authentication

- ServiceNow Login
 - Role-based Access Control
 - System Administrator Role
 - Session Authentication
 - Access Control Rule
-

9. User Interface

Include screenshots of:

- Home Page
- Incident Form
- Custom Fields
- Flow Designer

- Notification
 - Virtual Agent
 - Reports
 - Dashboard
-

10. Testing

Test Case	Expected Result
Create Incident	Success
Assign SLA	SLA starts
80% SLA	Notification received
Submit Justification	Saved successfully
Dashboard	Updated automatically

11. Screenshots / Demo

Include:

- Incident Creation
 - SLA Definition
 - Flow Designer
 - Notification
 - Virtual Agent (or documented conversation if licensing restricted)
 - Reports
 - Dashboard
 - GitHub Repository Link
 - Demo Video Link
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12. Known Issues

- Virtual Agent Designer requires additional licensing.
 - Flow Designer cannot directly launch Virtual Agent in the learning instance.
 - PDI limitations.
 - Some plugins unavailable in the training environment.
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13. Future Enhancements

- AI-based SLA Prediction
- Microsoft Teams Integration
- Slack Integration
- Email Notifications
- Mobile Push Notifications
- Performance Analytics
- Predictive Intelligence
- Automatic Manager Escalation

Conclusion:

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** provides an effective solution for improving Service Level Agreement (SLA) management within the ServiceNow platform. By automating SLA monitoring, sending proactive notifications to incident assignees, and capturing breach justifications through a structured workflow, the system reduces manual effort, improves accountability, and enhances operational transparency.

The project integrates key ServiceNow components such as Incident Management, Task SLA, Flow Designer, Notifications, Virtual Agent, Reports, and Dashboards to create an efficient and centralized SLA governance process. The generated reports and dashboards enable managers to monitor SLA performance, identify recurring breach patterns, and make informed decisions to improve service quality.

Although the training environment had limitations, including restricted access to certain Virtual Agent capabilities, the project successfully demonstrated the complete workflow and validated the proposed system design. The architecture is scalable and can be further enhanced in a production environment with advanced AI capabilities, predictive analytics, and integrations with collaboration platforms such as Microsoft Teams and Slack.